RJ45 / ETHERNET CABLE

A beginner-friendly guide for learners aged 18+

Learn the cable hardware, how network data travels, connector pinouts, wiring standards, crimping steps, testing, applications, and safety.

1. Learning goals

- Explain what an Ethernet cable does.
- Identify the cable, twisted pairs, jacket, modular plug, and contacts.
- Understand RJ45-style 8P8C connectors and T568A/T568B wiring.
- Describe a basic cable-making and testing workflow.
- Troubleshoot common wired-network problems safely.

2. What is an RJ45 cable?

In everyday networking, “RJ45 cable” usually means an Ethernet cable terminated with an 8-position, 8-contact (8P8C) modular plug. Strictly speaking, RJ45 is a registered-jack interface designation, while 8P8C describes the common plug shape. Most beginners and technicians still commonly say RJ45.

An Ethernet cable carries network signals between devices such as a computer, router, switch, access point, or network printer.

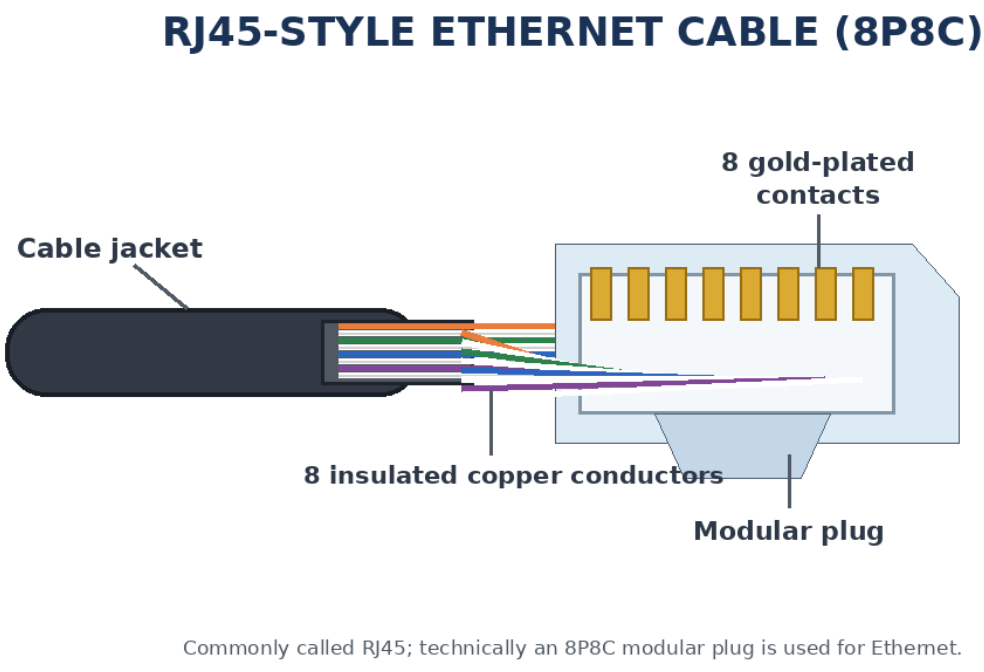


Figure 1. Simplified Ethernet cable and modular plug.

3. Hardware parts

Part	Description	Purpose
Outer jacket	Protective outer layer, often PVC or LSZH	Protects the internal conductors from wear.
Twisted pairs	Usually four pairs (eight conductors) in common copper Ethernet cable	Twisting helps reduce electromagnetic interference and crosstalk.
Copper conductors	Solid or stranded copper wires, depending on cable type	Carry electrical network signals.
Insulation	Thin insulation around each conductor	Keeps conductors separated.
8P8C modular plug	Plastic connector with eight contact positions	Connects the cable to a network port.
Gold-plated contacts	Metal contacts inside the plug	Make electrical contact with the port.
Boot / strain relief	Optional flexible sleeve near the plug	Helps reduce strain on the cable near the connector.

4. How does it work?

Step 1 — A device creates data: A computer or router prepares data to send over the network.

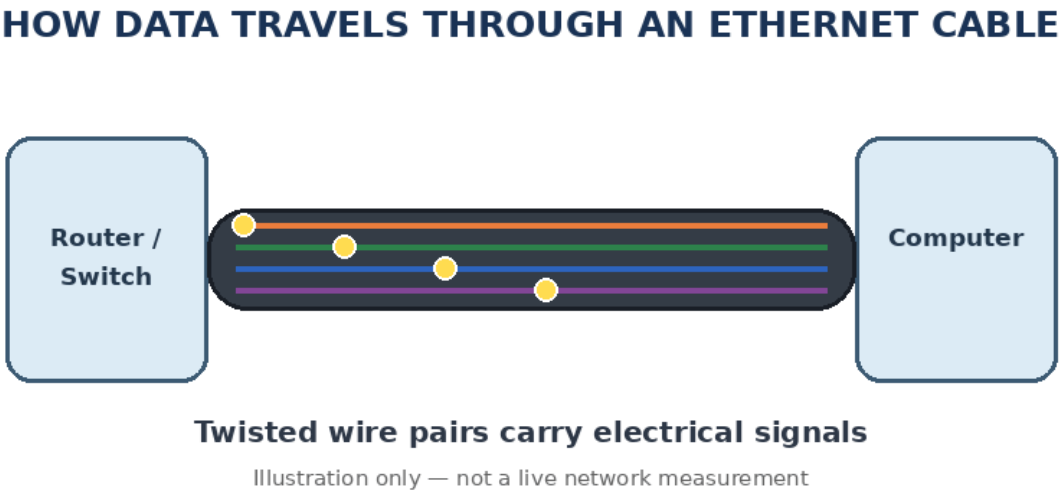
Step 2 — The network interface encodes signals: Ethernet hardware converts data into electrical signals for the cable.

Step 3 — Twisted pairs carry the signals: Signals travel through copper wire pairs; twisting helps limit interference.

Step 4 — The receiving device decodes the signals: The network interface at the other end interprets the signal and recovers the data.

Step 5 — Communication works both ways: Ethernet links can send and receive data; the details depend on the Ethernet standard and link speed.

Animated illustration: open “ethernet_data_flow.gif” in a browser or image viewer.



5. Cable categories and typical uses

Category	Common capability (typical maximum channel length)	Notes
Cat5e	1 Gb/s up to 100 m	Common for home and office Gigabit Ethernet.
Cat6	1 Gb/s up to 100 m; 10 Gb/s can be supported over shorter distances, commonly up to 55 m under suitable conditions	Performance depends on installation and crosstalk conditions.
Cat6A	10 Gb/s up to 100 m	Often chosen for new installations needing 10 Gigabit Ethernet.
Cat8	Designed for up to 25/40 Gb/s over up to 30 m	Primarily for short data-center links; not the usual home cable.

These are general standards-based examples, not guarantees for every cable or installation. Check the cable markings, equipment, and applicable Ethernet standard.

6. T568A and T568B wiring standards

T568A and T568B define the order of wire colors in the connector. For a normal straight-through patch cable, use the same standard at both ends. Both are valid when installed correctly; consistency is important.

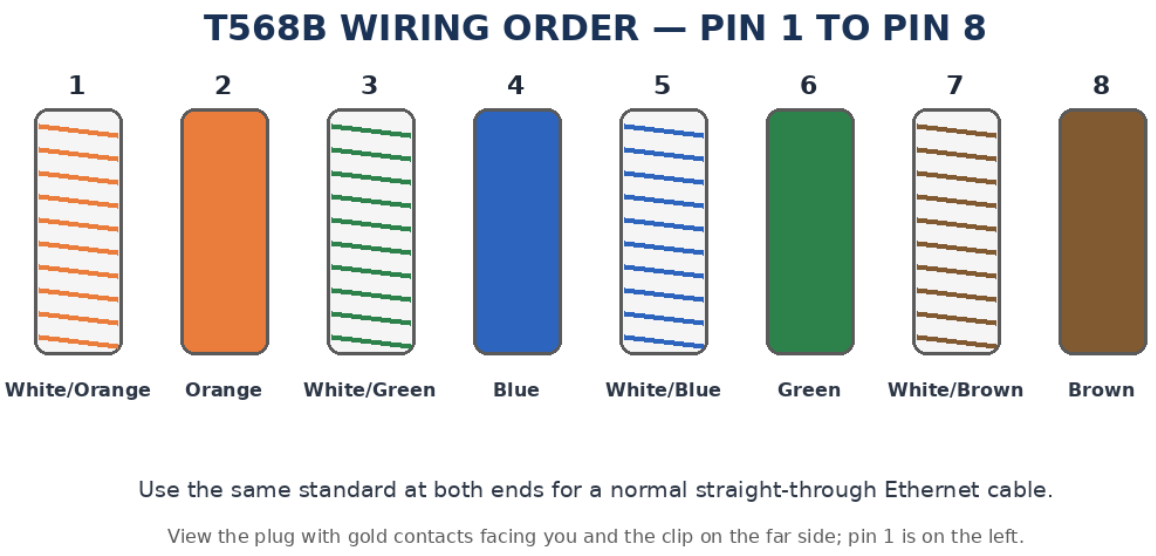


Figure 2. T568B color order, pin 1 through pin 8.

T568B pin order

- 1 — White/Orange
- 2 — Orange
- 3 — White/Green
- 4 — Blue
- 5 — White/Blue
- 6 — Green
- 7 — White/Brown
- 8 — Brown

Straight-through vs crossover

Straight-through: the same wiring standard (A-A or B-B) at both ends; the usual choice for patch cables. Crossover: T568A at one end and T568B at the other, historically used for certain like-device connections. Many modern ports support auto-MDI-X, so a crossover cable is often unnecessary.

7. Making an Ethernet patch cable

Typical tools: correctly rated Ethernet cable, compatible 8P8C plugs, cable stripper/cutter, crimping tool, and cable tester.

1. Choose the right cable category and connector type. Solid and stranded conductors may require different plug designs.
2. Cut the cable to the required length and carefully remove a short section of outer jacket without nicking the conductors.
3. Untwist only as much of each pair as needed and arrange the eight conductors in T568A or T568B order.
4. Trim the conductor ends evenly and insert them fully into the plug, checking that each wire reaches its contact channel and the jacket is held by the strain-relief area.
5. Crimp with a compatible tool.

6. Repeat at the other end using the same standard for a straight-through cable.
7. Test the cable for continuity, correct pin mapping, and shorts/splits if your tester supports them.

8. Testing and troubleshooting

Problem	Checks
No link light	Check both ends are fully inserted, test with a known-good cable, inspect ports and cable damage.
Link works at a lower speed	Check cable category and length, terminations, network port capabilities, and cable test results.
Intermittent connection	Inspect damaged clips, loose plugs, kinks, crushed cable, and poor crimps.
Tester shows open or short	Re-terminate the connector and check each conductor's pin mapping.
Only some pairs pass	Check wire order, split pairs, and the quality of both terminations.

9. Applications

- Connecting a PC to a router or switch
- Wired office and classroom networks
- Network printers and IP phones
- IP cameras and access points
- Some devices using Power over Ethernet (PoE)

PoE can carry electrical power as well as data on supported Ethernet cabling. Use compatible equipment and follow the equipment manufacturer's instructions.

10. Safety and good practice

- Never connect a cable or tools to exposed mains voltage.
- Do not look into or handle exposed conductors while equipment is powered if there is any risk from attached equipment.
- Use connectors and crimp tools designed for the cable type.
- Do not sharply bend, crush, staple through, or pull hard on Ethernet cable.
- Use outdoor-rated cable where exposed to weather and follow local building/fire requirements.
- Do not assume every cable supports PoE or every Ethernet speed; verify category, length, and equipment compatibility.

11. Quick quiz

1. What does an Ethernet cable connect?

2. What does 8P8C mean?

3. Why are conductors twisted into pairs?

4. Which wiring standard is shown in the diagram?

5. For a straight-through cable, what standard should be used at both ends?

6. Name two tools used to make and test a cable.

7. What might a cable tester report if one conductor is broken?

Answer key

1. Network devices such as a computer, router, or switch.

2. Eight positions, eight contacts.

3. To help reduce interference and crosstalk.

4. T568B.

5. The same standard at both ends: A-A or B-B.

6. A crimping tool and cable tester; also a stripper/cutter.

7. An open circuit or failed pin mapping.

12. Quick recap

An Ethernet cable carries network signals through copper conductors arranged as twisted pairs. The common connector is an 8P8C modular plug, often called RJ45. Correct pin order, quality terminations, suitable cable category, and testing are essential for a reliable link.

Identify the hardware → learn the pinout → terminate carefully → test before use.